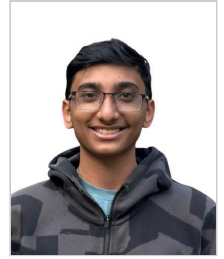


KUSH MEHTA

Curriculum Vitae

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Final-year Electronics and Communication Engineering student at Nirma University, applying for MS study beginning Fall 2027. Research interests span error-correcting codes and their hardware realization, computer architecture, and secure embedded systems. Recent work includes a real-time Galileo HAS decoder built at SAC/ISRO, a synthesizable $GF(2^8)$ arithmetic core verified against that decoder, and a co-authored paper on evaluation hazards in learned frequency-hopping receivers.

EDUCATION

Institute of Technology, Nirma University

B.Tech, Electronics & Communication Engineering — Minor in Data Science | CGPA 8.59 / 10

Ahmedabad, India

Aug 2023 – May 2027

- **Core coursework:** Digital Design, VLSI Design, Analog CMOS Integrated Circuits, Computer Architecture, Embedded Systems, Microcontrollers & Interfacing, Signals & Systems, Digital Signal Processing, Image Processing, Communication Systems, Computer Programming, Scripting Languages, Programming for Scientific Computing
- **Data Science minor:** Machine Learning, Statistics, Data Analysis & Visualization, Analytics of IoT, Big Data Systems

RESEARCH & PROFESSIONAL EXPERIENCE

Research Intern — Navigation Receiver Division

Space Applications Centre (SAC), ISRO · Real-Time Galileo High Accuracy Service (HAS) Decoding Utility · 10-week research internship

May – Jul 2026

Ahmedabad, India

- Built a real-time C++ utility that ingests a multiplexed TCP stream, tokenizes three interleaved protocols (RTCM payloads, Galileo HAS E6-B pages, Time-of-Transfer epoch tags), and routes each to an independent output port.
- Implemented Reed–Solomon (255,32) erasure decoding over $GF(2^8)$ from the Galileo HAS ICD, using lookup-table field arithmetic and Gaussian elimination for $k \times k$ matrix inversion, recovering complete messages despite page loss and out-of-order arrival.
- Reverse-engineered four undocumented Galileo RTCM 3.x message formats (1240–1243) and the CRC-24Q framing routine against an open-source implementation and the ICD, which unblocked the output stage.
- Synthesized RTCM 3.x output for GPS PRN 1–32 (messages 1057–1060) and Galileo PRN 1–36 (1240–1243) with correct framing and checksum.
- Delivered TCP streaming over WinSock 2.2 and Win32 serial on a pre-C++11 toolchain that ruled out modern STL containers and threading, using fixed-array structures, background-thread reads and stale-buffer purging.
- Validated end to end in RTKLIB at centimetre-level Precise Point Positioning, unit-testing every decoding stage against a Python reference implementation.

Supervisor: Sri. Pushkar Bhardwaj (Scientist/Engineer). Division Head: Smt. Saumi S. Faculty guide: Prof. Yogesh Trivedi.

Electronics Lead

Team Arrow UAV Club, Nirma University · Official aeromodelling and autonomous systems club

Sep 2023 – Present

Ahmedabad, India

- Lead a team of 8 to 12 across five competition campaigns, owning the full avionics stack from flight-controller bring-up to power design. Four podium finishes, including AIR 2 (SAE DDC Micro 2024), AIR 3 (2025) and 5th internationally on mission score (SAE Aero Design West 2025, USA).
- Trained 13 juniors into subsystem roles: three in piloting, five in autonomous systems and five in fixed-wing electric propulsion.
- Engineered the power plant for the 2025 international entry under a hard 400 W limit, matching motor and propeller on a thrust bench against analytical predictions. Designed the 2026 tilt-rotor VTOL to a 2:1 thrust-to-weight ratio.
- Designed autonomous ground-vehicle payload electronics (magnetometer, time-of-flight ranging, IR line-following) qualifying for the competition's highest 14× capture multiplier. Ran the scoring analysis that set team strategy: autonomous low-payload scores about 52 points against about 30 for manual high-payload.

RESEARCH

- **Simulation and Comparative Analysis of Lightweight Cryptography for Embedded Systems.** Research Methodology & Seminar, Nirma University, 2025–2026. Systematic comparison of AES-128-GCM, DES-CBC, ECC P-256 and ASCON-128 on throughput, peak memory and energy across 5,000 IoT payloads at 32B, 64B and 128B, with significance tested over 10 trial runs.

SELECTED PROJECTS

$GF(2^8)$ Arithmetic Core — Verilog, Exhaustively Verified

github.com/Kush-Git-379/gf256-rtl

2026

- Rebuilt the Galois-field arithmetic from the ISRO Reed–Solomon erasure decoder as synthesizable RTL over primitive polynomial 0x11D: two $GF(2^8)$ multiplier microarchitectures (log/antilog ROM and combinational shift-and-reduce), table inversion, and multiply-accumulate.

- Wrote self-checking SystemVerilog testbenches driven by golden vectors from the C++ original, covering all 65,536 multiplier pairs and 256 inverse cases. Proved the tests could fail by injecting three seeded faults.
- Quartus synthesis on Cyclone IV E made shift-and-reduce 13× smaller and 1.7× faster than the table version (62 LEs at 7.71 ns against 804 at 12.95 ns). Traced it to asynchronous ROM reads blocking M9K inference; registering the read address recovered block RAM and raised F_{max} to 180 MHz.

gem5 L1 Cache Sensitivity Analysis
Feb 2026

- Swept cache capacity, workload type and set associativity on gem5 v25.1 (X86 TimingSimpleCPU) to separate memory-bound from compute-bound behaviour.
- Growing 4 kB to 64 kB cut matrix-multiply miss rate from 5.06% to 0.04% (IPC +11.9%); 1-way to 8-way cut it from 4.74% to 0.34% (IPC +10.4%). An AES workload held flat at 0.14%, isolating execution units rather than memory as its bottleneck.

Cryptography Benchmark for Resource-Constrained Devices
Jan 2026

github.com/Kush-Git-379/crypto-algorithm-benchmark

- Benchmarked AES-GCM, DES, ECC P-256 and ASCON-128 across 5,000 IoT payloads, loading the ASCON C reference as a compiled DLL through Python ctypes for native-speed timing and profiling peak memory with tracemalloc. ASCON ran 78× faster than AES-GCM and 12× faster than DES.

Autonomous Tilt-Rotor VTOL Tricopter with AGV Payload
2025 – 2026

Electronics and avionics lead · SAE Aero Design East 2026

- Led the avionics stack for a hybrid tilt-rotor tricopter that takes off vertically, transitions to cruise and flies autonomously: TBS Lucid H7 controller with Here4 GPS for centimetre-level navigation, and a servo-actuated tilt mechanism with an airspeed-dependent interlock to prevent stall during transition.
- Validated the full mission in software-in-the-loop before first flight.

SecureCam — Edge AI Surveillance on Raspberry Pi
Apr 2026

Analytics of IoT · team of four

- Built a three-tier surveillance system on a Raspberry Pi 3B: Picamera2 capture over CSI, OpenCV MOG2 motion detection, authenticated MJPEG streaming over HTTP, SQLite session logging, and a Flask REST API serving a live dashboard.
- Exported YOLOv8n to ONNX and profiled it on the Pi; inference was slow enough to stall the video stream, so deep learning stayed on the laptop and the edge node ran MOG2 at 8–15 FPS instead. Concluded that real-time CNN inference on a Pi 3 needs either newer hardware or a dedicated accelerator.
- Sustained 2,256 s of continuous streaming across 39 sessions and 344 motion events at under 500 ms latency, with roughly 60 kB/hour of stored telemetry.

Secure GPS Telemetry Pipeline — ESP32, AES-128, MQTT
Mar 2026

github.com/Kush-Git-379/esp32-gps-encryption-tracker

- Built an end-to-end encrypted IoT pipeline: an ESP32 parses NMEA over UART2 with TinyGPS++, serializes to JSON, encrypts with AES-128 through mbedTLS, and publishes ciphertext to an MQTT broker on a 3-second cycle. A Python subscriber decrypts and resolves live position. Verified on hardware from GPS fix to decoded coordinates.

AWARDS & COMPETITIONS

- SAE Aero Design West 2025, Los Angeles, USA — 5th internationally on mission score. Electronics Lead.
- SAE India DDC Micro 2024 — All India Rank 2. Electronics Lead.
- SAE India DDC Micro 2025 — All India Rank 3. Electronics Lead.
- Technoxian 2024, RC Plane Challenge — 4th place.
- SAE Aero Design East 2026, USA and SAE India DDC Micro 2026 — Electronics Lead.

TECHNICAL SKILLS

Languages	C, C++, Verilog HDL, SystemVerilog, Python, Assembly, TCL, Bash, MATLAB, SQL
Digital Design & Verification	RTL design, functional verification, testbench development, golden-reference and exhaustive testing, fault injection, logic synthesis, static timing analysis, ModelSim, Quartus II, FPGA
Computer Architecture	gem5, microarchitecture, cache and memory hierarchy, set associativity, IPC and performance analysis, workload characterization, hardware/software co-design
Embedded & Firmware	ESP32, Raspberry Pi, Arduino, 8051, Pixhawk, ArduPilot / Mission Planner, Keil uVision, mbedTLS, bare-metal firmware, device drivers, UART / I ² C / SPI, real-time systems
Algorithms & Standards	Reed–Solomon, forward error correction, Galois Field GF(256), Galileo HAS ICD, RTCM 3.x, CRC-24Q, AES-128, ASCON-128, ECC P-256, DSP
Analog & Circuit Design	Cadence Virtuoso, LTspice, SPICE simulation, transient / DC / AC analysis, parametric sweep, pre- and post-layout simulation, MOSFET parameter extraction, current mirrors, CS / CD / CG, cascode and differential amplifier design
Data & ML	pandas, NumPy, Matplotlib, scikit-learn, PyTorch, OpenCV, KNIME, Tableau
Tools	Git, GitHub, Linux, RTKLIB, SolidWorks, Fusion 360, XFLR5